Theorem

Suppose that f is continuous on a finite closed interval $[a, b]$. If

$$\alpha = \inf_{a \le x \le b} f(x) \quad \text{and} \quad \beta = \sup_{a \le x \le b} f(x),$$

then α and β are respectively the minimum and maximum of f on $[a, b]$.

Namely, there exist $x_1, x_2 \in [a, b]$ with

$$f(x_1) = \alpha \quad \text{and} \quad f(x_2) = \beta.$$

Again, to prove this statement we will use compactness.

Proof: We will show that there exists $x_1 \in [a,b]$ such that $f(x_1) = \alpha$.

Suppose, by way of contradiction, that there exists no such x_1. In which case, $f(x) > \alpha$ for all $x \in [a,b]$.

Suppose that $t \in [a,b]$. Then $f(t) > \alpha$, and so

$$f(t) > \frac{f(t) + \alpha}{2} > \alpha$$

Since f is continuous at t there is an open interval I_t containing t such that for all $x \in I_t \cap [a,b]$

$$f(x) > \frac{f(t) + \alpha}{2}.$$

The collection $\mathcal{H} = \{ I_t \mid t \in [a,b] \}$ is an open cover of $[a,b]$.

Since $[a,b]$ is compact there exist $t_1, t_2 \dots t_n$ such that

$$\bigcup_{i=1}^{n} I_{t_i} \supseteq [a,b].$$

Set

$$\alpha_1 = \min_{i \in \{1,2,\dots n\}} \frac{f(t_i) + \alpha}{2}$$

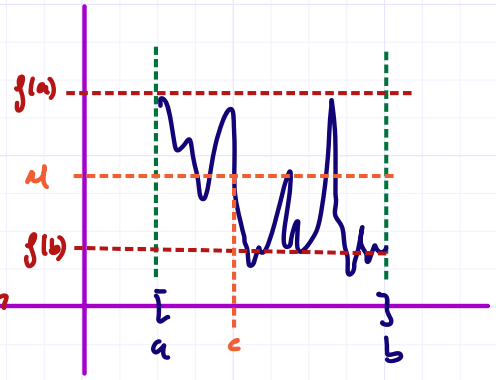
This implies that $\forall\ t \in [a,b]$

$$f(t) \ge \alpha_1 > \alpha \quad \Rightarrow \quad \lightning$$

∎

Theorem (Intermediate Value Theorem)

Suppose that f is continuous on $[a,b]$, $f(a) \neq f(b)$, and μ is between $f(a)$ and $f(b)$. Then $f(c) = \mu$ for some c in (a,b).



Proof: Suppose that $f(a) < \mu < f(b)$, and observe — the case when $f(b) < \mu < f(a)$ follows analogously

that the set

$$S = \{ x \mid x \in [a,b] \text{ and } f(x) \leq \mu \}$$

is bounded and non-empty.

Indeed $S \subseteq [a,b]$ and $a \in S$.

Let $c = \sup S$. To obtain the given statement,

we will show that $f(c) = \mu$.

If $f(c) > \mu$, then $c > a$ and since f is continuous

at c, there is an $\delta > 0$ such that if $x \in (c-\delta, c)$

$$f(x) > \mu.$$

in which case $c - \delta$ is an upper bound for S, yielding a

contradiction.

and so c is not an upper bound

for S, again yielding a contradiction.

Thus, $f(c) = \mu$. ∎

On the other hand if $f(c) < \mu$, then $c < b$ and since

f is continuous at c, there is a $\delta > 0$ such that

for all $x \in (c, c+\delta)$

$$f(x) < \mu$$

Uniform continuity

Definition

A function f is **uniformly continuous** on a subset S of its domain if, for every $\epsilon > 0$, there is a $\delta > 0$ such that, for all $x, x' \in S$ with $|x - x'| < \delta$

$$|f(x) - f(x')| < \epsilon.$$

We emphasise that in this definition δ depends only on ϵ and S and not on the particular choice of x and x', provided that they are both in S.

Example

- Consider the function $f : [-42, 423] \to \mathbb{R}$ be defined for $x \in [-42, 423]$ by

$$f(x) = x^2$$

Claim: f is uniformly continuous on $[-42, 423]$

Proof: Let $\varepsilon > 0$ be given and let $\delta \in (0, \varepsilon/84)$ for all $x, x' \in [-42, 423]$ with $|x - x'| < \delta$ we have

$$\begin{aligned} |f(x) - f(x')| &= |x^2 - x'^2| \\ &= |x - x'||x + x'| \\ &\leq 84\,|x - x'| \\ &< 84 \cdot \varepsilon/84 = \varepsilon \end{aligned}$$

Remark: In continuity at a point x_0, the δ can depend on x_0.

Uniform continuity asks for this not to be the case.

- **Claim:** The function $g: \mathbb{R} \to \mathbb{R}$ defined, for $x \in \mathbb{R}$, by $g(x) = x^2$ is not uniformly continuous on $\mathbb{R}$.

Proof: To prove this statement we will show that for any given $\delta > 0$ we can find $x, x' \in \mathbb{R}$ with $|x - x'| < \delta$ and $|g(x) - g(x')| > 1$.

Observe that for all $x, x' \in \mathbb{R}$

$$|g(x) - g(x')| = |x - x'||x + x'|$$

Let us choose $x, x' \in \mathbb{R}$ such that $x, x' > 1/\delta$ and $|x - x'| = \delta/2$

$$\Rightarrow |g(x) - g(x')| = |x - x'||x + x'| = \frac{\delta}{2}|x + x'|$$

$$> \frac{\delta}{2}\left(\frac{1}{\delta} + \frac{1}{\delta}\right) = 1$$

■

Theorem

If f is continuous on a closed and bounded interval $[a, b]$, then f is uniformly continuous on $[a, b]$.

Corollary

If f is continuous on a set T, then f is uniformly continuous on any finite closed interval contained in T.

To prove this result we will again appeal to the concept of compactness.

Proof of Thm:

Let $\varepsilon > 0$ be given,

By assumption f is continuous on $[a,b]$.

Thus for each $t \in [a,b]$ there exist $\delta_t > 0$

such that if $x \in [a,b]$ and $|x - t| < \delta_t$

$$|f(x) - f(t)| < \varepsilon/6$$

In this case, letting $I_t = (t - \delta_t, t + \delta_t)$

we observe that

$$\mathcal{H} = \{ I_t : t \in [a,b] \}$$

is an open cover of $[a,b]$. Since

$[a,b]$ is compact, by Heine-Borel's theorem

there exist $t_1, t_2 \dots t_n \in [a,b]$ with

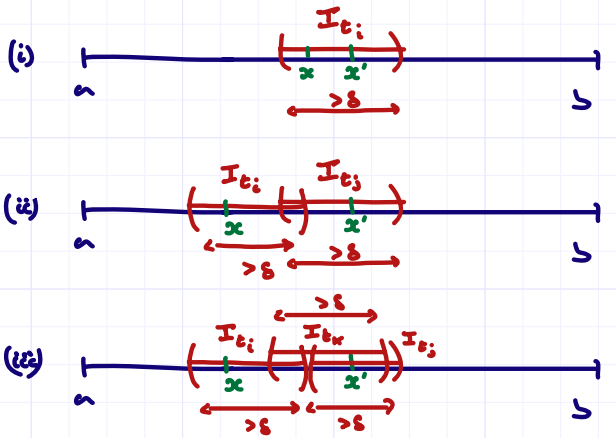
$$\bigcup_{i=1}^{n} I_{t_i} \supseteq [a,b].$$

Consider $\delta = \min \{ \delta_{t_1}, \delta_{t_2} \dots \delta_{t_n} \}$

We will see why this factor of 6 comes in later in the proof.

Let $x, x' \in [a,b]$ such $|x - x'| < \delta$

then we have one of three cases.



If case (i) occurs, then

$$|f(x) - f(x')| = |f(x) - f(t_i) + f(t_i) - f(x')|$$

$$< |f(x) - f(t_i)| + |f(t_i) - f(x')|$$

$$< \varepsilon/6 + \varepsilon/6 = 2\varepsilon/6 < \varepsilon/3 < \varepsilon.$$

If case (ii) occurs, let $y \in I_{t_i} \cap I_{t_j}$

and observe

$$|f(x) - f(x')| = |f(x) - f(t_i) + f(t_i) - f(y) + f(y) - f(t_j) + f(t_j) - f(x')|$$

$$\leq |f(x) - f(t_i)| + |f(t_i) - f(y)| + |f(y) - f(t_j)| + |f(t_j) - f(x')|$$

$$< \varepsilon/6 + \varepsilon/6 + \varepsilon/6 + \varepsilon/6 < \varepsilon.$$

Since $|x - t_i| < \delta_{t_i}$ $|y - t_i| < \delta_{t_i}$

$|y - t_j| < \delta_{t_j}$ $|x' - t_j| < \delta_{t_j}$

Case (iii): Left as an exercise.

■

Hint: Consider the claim.

$x \to t_i$ $t_i \to y$ $y \to t_k$ $t_k \to z$ $z \to t_j$ $t_j \to x'$

$y \in I_{t_i} \cap I_{t_k}$ $z \in I_{t_k} \cap I_{t_j}$

More about monotonic functions

Observe that if f is monotonic on an interval I, then f is either continuous or has a jump discontinuity at each $x \in I$. This and the intermediate value Theorem provide the key to the proof of the following theorem.

Theorem

Let $a, b \in \mathbb{R}$ with $a < b$. If f is real-valued, monotonic and non-constant on $[a, b]$, then f is continuous on $[a, b]$ if and only if its range $R_f = \{f(x) \mid x \in [a, b]\}$ is the closed interval whose boundary is the set $\{f(a), f(b)\}$.

Proof: We will prove the case when f is non-decreasing and leave the case when f is non-increasing as an exercise.

We first make some observations.

Since f is assumed to be non-decreasing we have that

$$\widetilde{R}_f = \{f(x) \mid x \in (a,b)\} \subseteq \left(\overbrace{\lim_{x \to a^+} f(x)}^{\alpha}, \overbrace{\lim_{x \to b^-} f(x)}^{\beta}\right)$$

which implies

$$R_f = \{f(x) \mid x \in [a,b]\} \subseteq \{f(a)\} \cup (\alpha, \beta) \cup \{f(b)\}$$

"$\Rightarrow$" Assume that f is continuous on $[a,b]$.

In which case $f(a) = \alpha$ and $f(b) = \beta$

Namely $R_f \subseteq [\alpha, \beta]$. Further, by the Intermediate Value Theorem.

given $\kappa \in [\alpha, \beta]$ then exists an $x_0 \in [a,b]$ such that $f(x_0) = \kappa$. Then, $R_f = [\alpha, \beta] = [f(a), f(b)]$.

"$\Leftarrow$" Suppose that $R_f = [f(a), f(b)]$. Since $f(a) \leq \alpha$ and $f(b) \geq \beta$ and since

$$R_f = \{f(x) \mid x \in [a,b]\} \subseteq \{f(a)\} \cup (\alpha, \beta) \cup \{f(b)\}$$

we have that $f(a) = \alpha$ and $f(b) = \beta$.

Since f is non-decreasing for $x_0 \in (a,b)$

$$\lim_{x \to x_0^-} f(x) \leq f(x_0) \leq \lim_{x \to x_0^+} f(x)$$

If either of these inequalities are strict, we would have that R_f would not be an interval.

Thus $f(x_0) = \lim_{x \to x_0} f(x)$ for all $x_0 \in (a,b)$.

All-in-all this implies that f is continuous. ∎

Theorem

Suppose that f is monotonic on an interval (a, b) and define

$$\alpha = \inf_{a<x<b} f(x) \quad \text{and} \quad \beta = \sup_{a<x<b} f(x).$$

(a) If f is nondecreasing, then

$$\lim_{x \to a^+} f(x) = \alpha \quad \text{and} \quad \lim_{x \to b^-} f(x) = \beta$$

(b) If f is non-increasing, then

$$\lim_{x \to a^+} f(x) = \beta \quad \text{and} \quad \lim_{x \to b^-} f(x) = \alpha$$

(c) If $a < x_0 < b$, then

$$\lim_{x \to x_0^+} f(x) \quad \text{and} \quad \lim_{x \to x_0^-} f(x)$$

exist and are finite. Moreover,

$$\lim_{x \to x_0^-} f(x) \leq f(x_0) \leq \lim_{x \to x_0^+} f(x)$$

if f is non-decreasing, and

$$\lim_{x \to x_0^-} f(x) \geq f(x_0) \geq \lim_{x \to x_0^+} f(x)$$

if f is non-increasing.

Theorem Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that f is real valued, strictly increasing and continuous on $[a, b]$, and let $f(a) = c$ and $f(b) = d$. There exists a unique function $g : [c, d] \to \mathbb{R}$, such that, for all $x \in [a, b]$ and all $y \in [c, d]$,

$$g(f(x)) = x \quad \text{and} \quad f(g(y)) = y.$$

Moreover, g is continuous and strictly increasing on $[c, d]$.

In other words f is invertible with continuous inverse.

Proof: We first show the existence of such a function g.

Since the hypothesis of our previous result is satisfied, we have that

$$R_f = \{f(x) : x \in [a,b]\} = [c,d]$$

that is for $y_0 \in [c,d]$ there exists $x_0 \in [a,b]$ with $f(x_0) = y_0$, and since f is strictly increasing there is only one such x_0. Thus we may define $g : [c,d] \to \mathbb{R}$ by $g(y_0) = x_0$.

In which case, by construction we have that $f(g(y_0)) = f(x_0) = y_0$ and $g(f(x_0)) = g(y_0) = x_0$.

Uniqueness of g.

The uniqueness of g follows from our assumption that f is strictly increasing.

Monotonicity of g.

To see that g is strictly increasing, suppose that $y_1, y_2 \in [c,d]$ with $y_1 < y_2$, and let $x_1, x_2 \in [a,b]$ be the unique points such that

$$f(x_1) = y_1 \quad \text{and} \quad f(x_2) = y_2$$

Since f is strictly increasing $x_1 < x_2$, thus

$$g(y_1) = x_1 < x_2 = g(y_2).$$

That is, g is strictly increasing.

Continuity of g.

Since, by construction

$$R_g = \{g(y) : y \in [c,d]\} = [a,b]$$

and since g is non-constant and increasing our previous result implies that g is continuous. ∎

2.3 Differentiable functions (single variable)

In calculus we study differentiation, emphasising rules for calculating derivatives. Here we consider the theoretical properties of differentiable functions, defining the derivative of a function in terms of limits.

Definition

A function f is **differentiable** at an interior point x_0 of its domain, if the quotient of differences, with $x \neq x_0$,

$$\frac{f(x) - f(x_0)}{x - x_0},$$

approaches a limit as x approaches x_0, in which case the limit is called the **derivative of f at x_0**, and is denoted by $f'(x_0)$. In other words,

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}.$$

It is sometimes convenient to let $x = x_0 + h$ and write the above equation as

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}.$$

If f is defined on an open set S, we say that f is **differentiable on S** if f is differentiable at every point of S. Observe that, if f is differentiable on S, then f' is a function on S.

We may also sometimes use the

familiar notation $\frac{d}{dx} f(x)$, or if we set

$y = f(x)$, we may use the notation $\frac{dy}{dx}$.

Note in this case y is implicitly

a function of x.

- We say that f is **continuously differentiable** on S if f' is continuous on S.
- If f is differentiable on a neighbourhood of x_0, it is reasonable to ask if f' is differentiable at x_0. If so, we denote the derivative of f' at x_0 by $f''(x_0)$. This is the **second derivative of f at x_0**, and it is also denoted by $f^{(2)}(x_0)$.
- Continuing inductively, if $f^{(n-1)}$ is defined on a neighbourhood of x_0, then the **n-th derivative of f at x_0**, denoted by $f^{(n)}(x_0)$, is the derivative of $f^{(n-1)}$ at x_0.
- For convenience we define the **zero-th derivative** of f to be f itself, namely $f^{(0)} = f$.

We may also sometimes use the term **Smooth** to mean that for all $n \in \mathbb{N}$, the n-th derivative exists.

Example

Let $n \in \mathbb{N}$ be fixed, and consider $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^n$.

Claim: $f'(x) = n x^{n-1}$

Proof of claim: Let $x_0 \in \mathbb{R}$, and observe that, for all $x \neq x_0$

$$\frac{f(x_0) - f(x)}{x_0 - x} = \frac{x_0^n - x^n}{x_0 - x} = \frac{(x_0 - x)\sum_{k=0}^{n-1} x^{n-k-1} x_0^k}{x_0 - x}$$

Using properties of telescoping sums.

$$= \sum_{k=0}^{n-1} x^{n-k-1} x_0^k$$

$$\Rightarrow \lim_{x \to x_0} \frac{f(x_0) - f(x)}{x_0 - x} = \lim_{x \to x_0} \sum_{k=0}^{n-1} x^{n-k-1} x_0^k = \sum_{k=0}^{n-1} \lim_{x \to x_0} x^{n-k-1} x_0^k = \sum_{k=0}^{n-1} x_0^{n-k-1} x_0^k$$

(Finite sum; Algebra of Limits.)

$$= \sum_{k=0}^{n-1} x_0^{n-1} = n x_0^{n-1}$$

∎

Interpretations of the derivative

- If $f(x)$ is the position of a particle at time $x \neq x_0$, the quotient of differences

$$\frac{f(x) - f(x_0)}{x - x_0}$$

is the average velocity of the particle between times x_0 and x. As x approaches x_0, the average applies to shorter and shorter intervals. Therefore, it makes sense to regard the limit

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0},$$

if it exists, as the particle's **instantaneous velocity at time** x_0.

This interpretation may be useful even if x is not time, in which case, we regard $f'(x_0)$ as the **instantaneous rate of change of** $f(x)$ **at** x_0, regardless of the specific nature of the variable x.

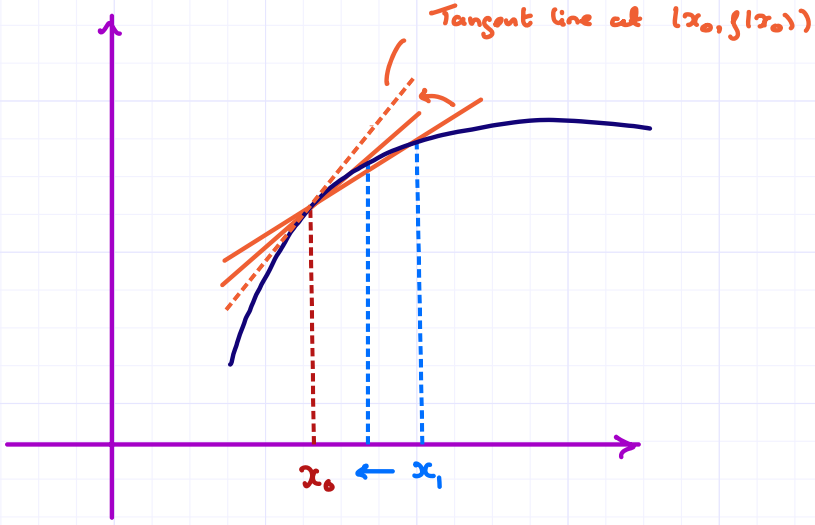
- The derivative also has a geometric interpretation. The equation of the line through two points $(x_0, f(x_0))$ and $(x_1, f(x_1))$ on the curve $y = f(x)$ is

$$y = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0).$$

Varying x_1 generates lines through $(x_0, f(x_0))$ that rotate into the line

$$y = f(x_0) + f'(x_0)(x - x_0)$$

as x_1 approaches x_0. This is the **tangent** to the curve $y = f(x)$ at the point $(x_0, f(x_0))$.



Properties of differentiable functions

Theorem

If f is differentiable at x_0, then f is continuous at x_0.

WARNING: The converse of this theorem is false, since a function may be continuous at a point without being differentiable at the point.

The function $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = |x|$ is continuous at $x = 0$ but not differentiable at $x = 0$.

Exercise: Prove this statement from first principles.

Proof of Thm: Let $\varepsilon > 0$ be given. Since by our hypothesis

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x_0) - f(x)}{x_0 - x}$$

exists, there exists a $\delta_1 > 0$ $(\delta_1 \in \mathbb{R})$ such that for all $x \in D_f$ with $|x - x_0| < \delta_1$,

$$|f'(x_0)| - \varepsilon \leq \left| \frac{f(x_0) - f(x)}{x_0 - x} \right| \leq |f'(x_0)| + \varepsilon$$

Setting $\delta = \min\left\{ \delta_1, \frac{\varepsilon}{|f'(x_0)| + \varepsilon} \right\}$. for all $x \in D_f$ with $|x - x_0| < \delta$, we have that

$$0 \leq |f(x_0) - f(x)| \leq (|f'(x_0)| + \varepsilon)|x_0 - x|$$

$$\leq (|f'(x_0)| + \varepsilon) \frac{\varepsilon}{|f'(x_0)| + \varepsilon} = \varepsilon$$

$$\Rightarrow \lim_{x \to x_0} f(x) = f(x_0)$$

$\Rightarrow f$ is continuous at x_0. ∎

Theorem

If f and g are differentiable at x_0, then $f+g$, $f-g$, and $f\cdot g$ are differentiable at x_0, with

- $(f+g)'(x_0) = f'(x_0) + g'(x_0)$;
- $(f-g)'(x_0) = f'(x_0) - g'(x_0)$;
- $(f\cdot g)'(x_0) = f'(x_0)\cdot g(x_0) + f(x_0)\cdot g'(x_0)$.

The quotient f/g is differentiable at x_0 if $g(x_0)\neq 0$, with

- $\left(\dfrac{f}{g}\right)'(x_0) = \dfrac{f'(x_0)\cdot g(x_0) - f(x_0)\cdot g'(x_0)}{(g(x_0))^2}$.

Proof: We will show the third statement and leave the other 3 as an exercise.

Observe that

$$\frac{f(x)g(x) - f(x_0)g(x_0)}{x-x_0} = \frac{f(x)g(x) - f(x_1)g(x) + f(x_0)g(x) - f(x_0)g(x_0)}{x-x_0}$$

$$= \frac{f(x)-f(x_1)}{x-x}\cdot g(x) + f(x_0)\,\frac{g(x)-g(x_0)}{x-x_0}$$

Thus, provided the limits exist

$$\lim_{x\to x_0}\frac{f(x)g(x) - f(x_1)g(x_1)}{x-x_0} = \lim_{x\to x_0}\frac{f(x)-f(x_0)}{x-x_0}\cdot \lim_{x\to x_0} g(x) + f(x_1)\cdot \lim_{x\to x_0}\frac{g(x)-g(x_1)}{x-x_0}$$

$$= f'(x_1)\cdot g(x_0) + f(x_1)g'(x_0).$$

Theorem (Chain rule)

Suppose that g is differentiable at x_0 and f is differentiable at $g(x_0)$. The composite function $h = f \circ g$, defined by

$$h(x) = f(g(x)),$$

is differentiable at x_0, with

$$h'(x_0) = f'(g(x_0))g'(x_0).$$

Pf: If we set $v = g(x)$, the statement we are asked to prove becomes, short

$$\frac{d}{dx} f(v) = f'(v) \frac{dv}{dx} \quad \text{at } x = x_0$$

Since $v = g(x)$, v is a function of x and v is diffble wrt respect to x, hence

$$\lim_{h\to 0} \left(\frac{v(x_0+h) - v(x_0)}{h} - v'(x_0) \right) = \left(\lim_{h\to 0} \frac{v(x_0+h) - v(x_0)}{h} \right) - v'(x_0)$$

$$= v'(x_0) - v'(x_0) = 0.$$

For h small, we def

$$V(h) = \begin{cases} \dfrac{v(x_0+h) - v(x_0)}{h} - v'(x_0) & \text{if } h \neq 0 \\ 0 & \text{if } h = 0 \end{cases}$$

and note that by our above argument V is continuous at $h = 0$.

Thus,

$$v(x_0+h) = v(x_0) + h(V(h) + v'(x_0))$$

Since we also know that f is diffble at $z_0 = g(x_0)$, setting

$$U(k) = \begin{cases} \dfrac{f(z_0+k) - f(z_0)}{k} - f'(z_0) & \text{if } k \neq 0 \\ 0 & \text{if } k = 0 \end{cases}$$

we also observe that U is continuous at $k = 0$, and

$$f(z_0+k) = f(z_0) + k(U(k) + f'(z_0)).$$

Next we recall that

$$\frac{d}{dx} f(v(x_0)) = \lim_{h\to 0} \frac{f(v(x_0+h)) - f(v(x_0))}{h}$$

Note the position of h.

Substitute.

$$\frac{d}{dx} f(v(x_0)) = \lim_{h\to 0} \frac{f\big(\underbrace{v(x_0)}_{z_0} + \underbrace{h(v(h)+v'(x_0))}_{\text{view this as } k} - f(\underbrace{v(x_0)}_{z_0})\big)}{h}$$

$$= \lim_{h\to 0} \frac{f(\underbrace{v(x_0)}_{z_0}) + \underbrace{h(v(h)+v'(x_0))}_{k}\Big(\underbrace{v\big(\underbrace{h(v(h)+v'(x_0))}_{k} + f'(\underbrace{v(x_0)}_{z_0})\big) - f(\underbrace{v(x_0)}_{z_0})}_{z_0}\Big)}{h}$$

$$= \lim_{h\to 0} \frac{h(v(h)+v'(x_0))\big(v(k) + f'(v(x_0))\big)}{h}$$

(i.e. k is a function of h.)

$$= \lim_{h\to 0} \underbrace{(v(h)+v'(x_0))}_{\text{converges to } v'(x_0) \text{ as } h\to 0}\big(v(k) + \underbrace{f'(v(x_0))}_{\text{converges to } f'(v(x_0)) \text{ as } h\to 0}\big)$$

$$= v'(x_0)\cdot f'(v(x_0)) = f'(v(x_0))\cdot \frac{dv}{dx}$$

Recall: $k = h(v(h)+v'(x_0))$

and so $\lim_{h\to 0} k = \lim_{h\to 0} h(v(h)+v'(x_0)) = 0$

$\Rightarrow \lim_{h\to 0} v(k) = v(\lim_{h\to 0} k) = v(0) = 0$

Since v is continuous at $k=0$

■

Definition

- If f is a function defined on an interval $[x_0, b)$, the **right-hand derivative of f at x_0** is defined to be

$$f'_+(x_0) = \lim_{x \to x_0+} \frac{f(x) - f(x_0)}{x - x_0}$$

  if the limit exists.

- If f is a function is defined on an interval $(a, x_0]$, the **left-hand derivative of f at x_0** is defined to be

$$f'_-(x_0) = \lim_{x \to x_0-} \frac{f(x) - f(x_0)}{x - x_0}$$

  if the limit exists.

- We say that f is **differentiable on the closed interval** $[a, b]$ if f is differentiable on the open interval (a, b) and $f'_+(a)$ and $f'_-(b)$ both exist.

- We say that f is **continuously differentiable on** $[a, b]$ if f is differentiable on $[a, b]$, f' is continuous on (a, b) with

$$f'_+(a) = \lim_{x \to a^+} f'(x) \quad \text{and} \quad f'_-(b) = \lim_{x \to b^-} f'(x).$$

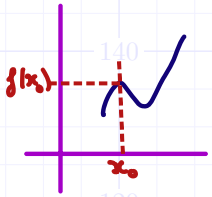
Extreme Values

Definition

Let f denote a real-valued function. We say that $f(x_0)$ is a **local extreme value** of f if there is a $\delta > 0$ such that $f(x) - f(x_0)$ does not change sign on $(x_0 - \delta, x_0 + \delta) \cap D_f$.

- $f(x_0)$ is a **local maximum value** of f if

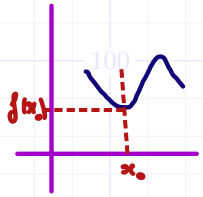
$$f(x) \leq f(x_0)$$



for all $x \in (x_0 - \delta, x_0 + \delta) \cap D_f$. In which case, the point x_0 is called a **local extreme point** of f, or, more specifically, a **local maximum point** of f.

- $f(x_0)$ is a **local minimum value** of f if

$$f(x) \geq f(x_0)$$



for all $x \in (x_0 - \delta, x_0 + \delta) \cap D_f$. In which case, the point x_0 is called a **local extreme point** of f, or, more specifically, a **local minimum point** of f.

Theorem

If f is differentiable at a local extreme point $x_0 \in D_f^0$, then $f'(x_0) = 0$.

If $f'(x_0) = 0$, we say that x_0 is a **critical point** of f. The above Theorem implies that every local extreme point of f at which f is differentiable is a critical point of f. The converse is false. For example, 0 is a critical point of $f(x) = x^3$, but not a local extreme point.

Proof. We will show that if f is diffble at x_0 and if $x_0 \in D_f^\circ$ is not a local extreme point of f then $f'(x_0) \neq 0$.

We will use a similar proof as in the chain rule.

To this end for $x \in D_f$ in an open neighborhood of x_0, set

$$E(x) = \begin{cases} \dfrac{f(x) - f(x_0)}{x - x_0} - f'(x_0) & \text{if} \\ 0 & \text{if} \end{cases}$$

and note that $\lim_{x \to x_0} E(x) = 0$

$$f'(x_0) + E(x) = \frac{f(x) - f(x_0)}{x - x_0} \quad \text{if } x \neq x_0.$$

If $f'(x_0) \neq 0$, then there exists $\delta > 0$ such that for all $x \in D_f$ with $|x - x_0| < \delta$.

$$|E(x)| < |f'(x)|$$

Thus, either

$$\frac{f(x) - f(x_0)}{x - x_0} > 0 \quad \text{for all } x \text{ in a neighbor of } x_0$$

or

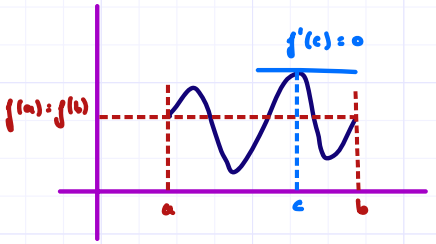
$$\frac{f(x) - f(x_0)}{x - x_0} < 0 \quad \text{for all } x \text{ in a neighbor of } x_0$$

Here. Since $x - x_0$ changes sign in every neighbor of x_0, $f(x) - f(x_0)$ must change sign, i.e. x_0 is not an extreme point of f. ∎

Rolle's Theorem

Theorem (Rolle's Theorem)

Let $a, b \in \mathbb{R}$ with $a < b$ be fixed. Suppose that f is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b), and $f(a) = f(b)$. Then $f'(c) = 0$ for some c in the open interval (a, b).



Proof: Since f is continuous on $[a,b]$, f attains its maximum and minimum. If these two extreme values are the same, then f is constant on (a,b) and so $f'(x) = 0$ $\forall$ all $x \in (a,b)$.

On the other hand if the extreme values are different, then at least one must be obtained in the open interval (a,b) at some point c and hence by our previous theorem we have $f'(c) = 0$. ∎

} Its result is sometimes referred to as the Extreme Value Thm.

Theorem (Intermediate value theorem for derivatives)

Let $a, b \in \mathbb{R}$ with $a < b$ be fixed. Suppose that f is differentiable on $[a, b]$, that $f'_+(a) \neq f'_-(b)$, and that μ is between $f'_+(a)$ and $f'_-(b)$. Then $f'(c) = \mu$ for some c in (a, b).

A derivative of a function f may exist on an interval $[a,b]$ without being continuous. Nevertheless, a version of the intermediate value theorem holds for derivatives.

Proof: Suppose, without loss of generality,

$$f'_+(a) < \mu < f'_-(b)$$

and define $g: [a,b] \to \mathbb{R}$, by

$$g(x) = f(x) - \mu x \quad \text{for all } x \in [a,b]$$

in which case, for $a \leq x \leq b$

$$g'(x) = f'(x) - \mu$$

where we look at left and right derivatives at b and a respectively.

Thus,

$$g'(a) < 0 \quad \text{and} \quad g'(b) > 0 \qquad (*)$$

Since g is continuous on $[a,b]$, it attains a minimum at some point $c \in [a,b]$.

By (*) and the definition of the right and left derivatives, there exists a $\delta > 0$ such that for all $x \in (a, a+\delta)$ and $y \in (b-\delta, b)$,

$$g(x) < g(a) \quad \text{and} \quad g(y) < g(b)$$

Hence, $c \neq a$ and $c \neq b$.

Since c is an extreme point

Thus, there exists $c \in (a,b)$ such that $g'(c) = 0$

and hence $f'(x) = \mu$. ∎

Theorem (Generalised mean value theorem)

Let $a, b \in \mathbb{R}$ with $a < b$ be fixed. If f and g are continuous functions on the closed interval $[a, b]$ and differentiable on the open interval (a, b), then

$$(g(b) - g(a))f'(c) = (f(b) - f(a))g'(c)$$

for some c in (a, b).

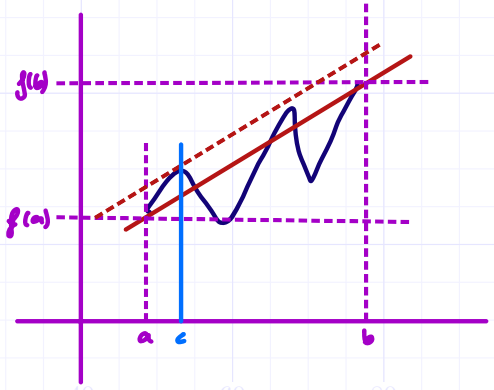
Theorem (Mean value theorem)

Let $a, b \in \mathbb{R}$ with $a < b$ be fixed. If f is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b), then

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

for some c in (a, b).

Proof: Follows from the Generalised mean value theorem taking $g(x) = x$. ∎



Proof of Generalised mean value Thm:

Observe that the function $h: [a,b] \to \mathbb{R}$ defined, for $x \in [a,b]$, by

$$h(x) = (g(b) - g(a))f(x) - (f(b) - f(a))g(x).$$

is continuous on $[a,b]$, and differentiable on (a,b), and

$$h(a) = h(b) = g(b)f(a) - f(b)g(a)$$

Therefore, Rolle's theorem implies that, $h'(c) = 0$ for some $c \in (a,b)$

Since

$$h'(c) = (g(b) - g(a))f'(c) - (f(b) - f(a))g'(c).$$

the results follows. ∎

Consequences of the mean value theorem

Theorem

If $f'(x) = 0$ for all x in (a, b), then f is constant on (a, b).

Theorem

If f' exists and does not change sign on (a, b), then f is monotonic on (a, b).

Theorem

If, for some $M \in \mathbb{R}$,

$$|f'(x)| \leq M, \quad a < x < b,$$

then

$$|f(x) - f(x')| \leq M|x - x'|, \checkmark\ x, x' \in (a, b).$$

A function satisfying this property is sometimes called Lipschitz.

3 – Infinite Sequences and Series

Sequences and series are the workhorse of analysis and forms the foundations of most mathematical applications they appear, for instance, in signal analysis (Fourier and Taylor series) and physics (general harmonic motion).

An **infinite sequence** (more briefly, a **sequence**), of real numbers is a real-valued function defined on a set of integers $\{n \in \mathbb{Z} \mid n \geq k\}$, for some $k \in \mathbb{Z}$. We call the values of the function the **terms** of the sequence. We denote a sequence by listing its terms in order; thus,

$$(s_n)_{n=k}^{\infty} = (s_k, s_{k+1}, \dots).$$

When $k = 1$, we may sometimes write $(s_n)_{n\in\mathbb{N}}$ for $(s_n)_{n=1}^{\infty}$.

Important note: A sequence and a set are different. Indeed, a set may not be ordered, and a set does not contain repeated elements.

To highlight this, unlike in the lecture notes, we will use the notation $(s_n)_{n=k}^{\infty}$ over the notation $\{s_n\}_{n=k}^{\infty}$ – However, you may use which one you find most natural.

Example

$\left(\frac{1}{n^2+1}\right)_{n=0}^{\infty} = \left(1, \frac{1}{2}, \frac{1}{5}, \frac{1}{10}, \dots, \frac{1}{n^2+1}, \dots\right)$

$\left((-1)^n\right)_{n=0}^{\infty} = (1, -1, 1, -1, \dots)$

$\left(\frac{1}{n-2}\right)_{n=3}^{\infty} = \left(1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}, \dots\right)$

Note: The sequence $\left((-1)^n\right)_{n=0}^{\infty}$ and the set $\{1, -1, 1, -1, \dots\} = \{-1, 1\}$ are not the same.

Given a sequence $(s_n)_{n=k}^{\infty} = (s_k, s_{k+1}, \dots)$, as above, for $n \in \mathbb{N}$, the real number s_n is the $(n-k+1)$**-th term** of the sequence.

Note, two sequences comprising of the same terms but with an index shift are considered to be identical, for instance

$$\left(\frac{1}{n-2}\right)_{n=3}^{\infty} \quad \text{and} \quad \left(\frac{1}{n}\right)_{n=1}^{\infty}$$

are considered to be identical.

The interesting questions about a sequence $(s_n)_{n=k}^{\infty}$ of real numbers concern the behaviour of s_n for large n.

Definition (Convergent sequences)

A sequence $(s_n)_{n=k}^{\infty}$ of real numbers **converges to a limit** s if for every $\epsilon > 0$ there exists an integer N such that, for all integers $n > N$,

$$|s_n - s| < \epsilon$$

In this case, we say that $(s_n)_{n=k}^{\infty}$ is **convergent** and write

$$\lim_{n\to\infty} s_n = s.$$

A sequence that does not converge is called **divergent**.

1st term, 2nd term

$\left(\frac{1}{n-2}\right)_{n=3}^{\infty} = \left(1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}, \dots\right)$

↑ 3rd term.

Note: N can be dependent on ϵ.

Example

- Let $c \in \mathbb{R}$ and consider the sequence $(s_k)_{k=1}^{\infty}$ given by $s_k = c$ for all $k \in \mathbb{N}$.

  Claim: $\lim_{k \to \infty} s_k = c$

  Proof: Let $\varepsilon > 0$ be given and let $N \in \mathbb{N}$

  For all integers $k \geq N$ we have that

  $|s_k - c| = |c - c| = 0 < \varepsilon$

  Thus $(s_k)_{k=1}^{\infty}$ is a convergent sequence and converges to the limit c.

- Let $(s_n)_{n \in \mathbb{N}}$ be the sequence defined by

  $$s_n = \frac{2n+1}{n+1}$$

  for $n \in \mathbb{N}$.

  Claim: $\lim_{n \to \infty} s_n = 2$

  Proof: Let $\varepsilon > 0$ be given, and let $N \in \mathbb{N}$ be such that $N > 1/\varepsilon$.

  For all integers $n \geq N$, we have that

  $$|s_n - 2| = \left| \frac{2n+1}{n+1} - 2 \right|$$

  $$= \left| \frac{2n+1}{n+1} - \frac{2n+2}{n+1} \right|$$

  $$= \frac{1}{n+1} \leq \frac{1}{N} < \varepsilon$$

  Thus $(s_n)_{n \in \mathbb{N}}$ is a convergent sequence and converges to the limit 2.

We say that the terms of a sequence $(s_n)_{n=k}^{\infty}$ satisfy a given condition for all n if s_n satisfies the condition for all integers $n \geq k$, or for large n if there is an integer $N > k$ such that s_n satisfies the condition whenever $n \geq N$.

For example, the terms of the sequnce $(1/n)_{n=1}^{\infty}$ are positive for all n, while those of $(1 - 7/n)_{n=1}^{\infty}$ are positive for $n > N = 8$.

Definition (Sequences diverging to infinity)

Let $(s_n)_{n=1}^{\infty}$ denote a sequence of real numbers. We write

$$\lim_{n\to\infty} s_n = \infty$$

if for any real number a, there exists an $N \in \mathbb{N}$, such that $s_n > a$ for all integers $n \geq N$, in which case we say that $(s_n)_{n=k}^{\infty}$ **diverges to** ∞. Similarly, we write

$$\lim_{n\to\infty} s_n = -\infty$$

if for any real number a, there exists an $N \in \mathbb{N}$, such that $s_n < a$ for all integers $n \geq N$, in which case we say that $(s_n)_{n=k}^{\infty}$ **diverges to** $-\infty$.

Example

- Consider the Sequence $\left(\frac{n}{2} + \frac{1}{n}\right)_{n\in\mathbb{N}}$

claim: The Sequence diverges to infinity.

Proof: Observe that if a is any positive real number, then let $N \in \mathbb{N}$ with $N \geq 2a$ we have, for all integers $n \geq N$,

$$\frac{n}{2} + \frac{1}{n} \geq \frac{N}{2} + 0 \geq \frac{2a}{2} = a.$$

- Consider the Sequence $(n - n^2)_{n\in\mathbb{N}}$

claim: The Sequence diverges to infinity.

Proof: Observe that given $a \in \mathbb{R}$, then letting $N \in \mathbb{N}$ with $N > \sqrt{|a|} + 1$ for all integers $n \geq N$,

$$n - n^2 = n(n-1) \leq -(n-1)^2 \leq -(N-1)^2 < -|a| \leq a$$

Our next result enables us to apply the theory of limits we have developed for function to some sequences.

Theorem

Let $\lim_{x\to\infty} f(x) = L$, where L is in the extended reals, and suppose that $s_n = f(n)$ for $n \in \mathbb{N}$ large. Then

$$\lim_{n\to\infty} s_n = L.$$

Example

- Consider the sequence $(s_n)_{n\in\mathbb{N}}$ where

$$s_n = \frac{\log(n)}{n}$$

and consider the function $g:(0,\infty)\to\mathbb{R}$ defined by $g(x) = \frac{\log(x)}{x}$.

Observe that $g(n) = s_n$ for all $n\in\mathbb{N}$ and that by L'Hôpital's rule

$$\lim_{x\to\infty} g(x) = \lim_{x\to\infty} \frac{\log(x)}{x} = \lim_{x\to\infty} \frac{1/x}{1} = 0$$

Hence, we have

$$\lim_{n\to\infty} \frac{\log(n)}{n} = \lim_{x\to\infty} g(x) = 0.$$

Note: only applies to functions.

namely if $\frac{h(x_0)}{g(x_0)}$ is of an indeterminant form and if h and g are differentiable at x_0, then

$$\lim_{x\to x_0} \frac{h(x)}{g(x)} = \lim_{x\to x_0} \frac{h'(x)}{g'(x)}.$$

Where differentiable

Because differentiable

And of indeterminant form.

Definition (Subsequence)

A sequence $(t_m)_{m=l}^{\infty}$ is a **subsequence** of a sequence $(s_n)_{n=k}^{\infty}$ if

$$t_m = s_{n_m}$$

where $(n_m)_{m=l}^{\infty}$ is ~~an~~ a strictly increasing infinite sequence of integers, with $n_l \geq k$.

We often write the subsequence $(t_k)_{k=l}^{\infty}$ by $(s_{n_m})_{m=l}^{\infty}$.

By strictly increasing we mean that $n_{m+1} > n_m$ for all $m \in \mathbb{N}$.

Note that $(s_n)_{n=k}^{\infty}$ is a subsequence of itself, as can be seen by taking $n_m = m$. All other subsequences of $(s_n)_{n=k}^{\infty}$ are obtained by deleting terms from $(s_n)_{n=k}^{\infty}$ and leaving those remaining in their original relative order.

Example

- If $(s_n)_{n\in\mathbb{N}} = (1/n)_{n\in\mathbb{N}} = (1, 1/2, 1/3, 1/4, \dots)$

  then with $n_m = 2m$ for all $m \in \mathbb{N}$.

  we have that

  $$(s_{2m})_{m\in\mathbb{N}} = (1/2m)_{m\in\mathbb{N}} = (1/2, 1/4, 1/6, \dots, 1/2m, \dots)$$

  is a subsequence of $(s_n)_{n\in\mathbb{N}}$.

  Letting $n_m = 2m+1$ for all $m \in \mathbb{N}$

  we have that

  $$(s_{2m+1})_{m\in\mathbb{N}} = (1/2m+1)_{m\in\mathbb{N}} = (1/3, 1/5, \dots, 1/2m+1, \dots)$$

  is a subsequence of $(s_n)_{n\in\mathbb{N}}$.

- The sequence $(s_n)_{n\in\mathbb{N}} = \left((-1)^n\left(1+\frac{1}{n}\right)\right)_{n\in\mathbb{N}}$

does not converge. Exercise: Try proving this.

But does have subsequences which converge.

For example

$$(s_{4n})_{n\in\mathbb{N}} = \left(1+\frac{1}{4n}\right)_{n\in\mathbb{N}}$$

Indeed, $\lim_{n\to\infty} s_{4n} = \lim_{n\to\infty} 1+\frac{1}{4n} = 1$ Exercise: Try proving this yourself.

On the other hand,

$$(s_{4n+3})_{n\in\mathbb{N}} = \left(-\left(1+\frac{1}{4n+2}\right)\right)_{n\in\mathbb{N}}$$

and

$$\lim_{n\to\infty} s_{4n+3} = \lim_{n\to\infty} -\left(1+\frac{1}{4n+3}\right) = -1$$ Exercise: Try proving this yourself.

Theorem

Let $(s_n)_{n=1}^{\infty}$ denote a sequence of real numbers. If there exists an $s \in \mathbb{R}$, with

$$\lim_{n\to\infty} s_n = s,$$

then

$$\lim_{m\to\infty} s_{n_m} = s$$

for every subsequence $(s_{n_m})_{m=l}^{\infty}$ of $(s_n)_{n=1}^{\infty}$.

Exercise: Does a similar result hold if the seq. diverges to ∞ or if the seq. diverges to $-\infty$.

Proof: If we know that $\lim_{n\to\infty} s_n = s$, then given a $\varepsilon > 0$ $\exists\, N \in \mathbb{N}$ such that for all integers $n \geq N$

$$|s_n - s| < \varepsilon$$

Since $(n_m)_{m=l}^{\infty}$ is an increasing seq. then exists $M \in \mathbb{N}$ such that, $n_m \geq N$ for all integers $m \geq M$. Therefore, for all $m \geq M$, $n_m \geq N$ and so

$$|s_{n_m} - s| < \varepsilon$$

Yielding $\lim_{m\to\infty} s_{n_m} = s$.

■

For a given set S we defined **limit points** in terms of neighborhoods: $\overline{x}$ is a limit point of a set S if every neighborhood of $\overline{x}$ contains points of S distinct from $\overline{x}$. Our next result shows that an equivalent definition can be stated in terms of sequences.

Theorem

A point $\overline{x}$ is a limit point of a set S if and only if there exists a sequence $(x_n)_{n\in\mathbb{N}}$ of points in S such that $x_n \neq \overline{x}$ for all $n \in \mathbb{N}$, and

$$\lim_{n\to\infty} x_n = \overline{x}.$$

Proof: "=>" Let $\bar{x}$ be a limit point of S.

Then for every $n \in \mathbb{N}$ the interval $(\bar{x} - 1/n, \bar{x} + 1/n)$ contains a point $x_n \neq \bar{x}$ with $x_n \in S$. Now given an $\varepsilon > 0$ with $N \in \mathbb{N}$ with $N > 1/\varepsilon$, we have that for all $n \geq N$, that

$$|x_n - \bar{x}| \leq 1/n \leq 1/N < \varepsilon.$$

Thus $\lim_{n\to\infty} x_n = \bar{x}$.

"<=" Suppose that there exists a sequence $(x_n)_{n\in\mathbb{N}}$ with $x_n \in S$ with $x_n \neq \bar{x}$ for all $n \in \mathbb{N}$ $\lim_{n\to\infty} x_n = \bar{x}$. With $\varepsilon > 0$ all there exists $n \in \mathbb{N}$ such that

$$|x_n - \bar{x}| < \varepsilon$$

which implies that $x_n \in (\bar{x} - \varepsilon, \bar{x} + \varepsilon)$

Moreover, $x_n \neq \bar{x}$ all that

$(\bar{x} - \varepsilon, \bar{x} + \varepsilon) \cap S \setminus \{\bar{x}\} \supseteq \{x_n\}$

Non-empty.

$\Rightarrow$ $\bar{x}$ is a limit point of S. ∎

Theorem

(a) If $(x_n)_{\mathbb{N}}$ is bounded sequence of real numbers, then $(x_n)_{n\in\mathbb{N}}$ has a convergent subsequence.

(b) If $(x_n)_{n\in\mathbb{N}}$ is a sequence of real numbers unbounded above, then $(x_n)_{n\in\mathbb{N}}$ has a subsequence $(x_{n_k})_{k\in\mathbb{N}}$ such that

$$\lim_{k\to\infty} x_{n_k} = \infty.$$

(c) If $(x_n)_{n\in\mathbb{N}}$ is a sequence of real numbers unbounded below, then $(x_n)_{n\in\mathbb{N}}$ has a subsequence $(x_{n_k})_{k\in\mathbb{N}}$ such that

$$\lim_{k\to\infty} x_{n_k} = -\infty.$$

} This statement is in fact equivalent to the Bolzano-Weierstrass Theorem

Every bounded infinite set of real numbers has at least one limit point.

Proof: We will prove Part (a) and leave (b) and (c) as an exercise.

Proof of (a): Let S be the set of distinct numbers that occur as terms of the sequence $(x_n)_{n\in\mathbb{N}}$

If S contains only finitely many points then some $\bar{x}$ in S occurs infinitely often in the seq. $(x_n)_{n\in\mathbb{N}}$; that is $(x_n)_{n\in\mathbb{N}}$ has a subsequence $(x_{n_k})_{k\in\mathbb{N}}$ such that $x_{n_k} = \bar{x}$ for all $k \in \mathbb{N}$.

Then,

$$\lim_{k\to\infty} x_{n_k} = \bar{x}.$$

e.g. If $(x_n)_{n\in\mathbb{N}}$ was the seq. $((-1)^n)_{n\in\mathbb{N}}$ then $S = \{-1, 1\}$

If $(x_n)_{n\in\mathbb{N}}$ was the seq. $(1,2,3,1,2,3,1,2,3,\dots)$ then $S = \{1,2,3\}$.

If $(x_n)_{n\in\mathbb{N}}$ was the seq. $(1/n)_{n\in\mathbb{N}}$ then $S = \{\frac{1}{n} \mid n \in \mathbb{N}\} = \{1, \frac{1}{2}, \frac{1}{3}, \dots\}$.

If S is infinite, since $(x_n)_{n\in\mathbb{N}}$ is a bounded sequence, we know that the set S is bounded. } namely $\exists M \in \mathbb{N}$ such that for all $n \in \mathbb{N}$ $|x_n| \leq M$.

The Bolzano-Weierstrass theorem implies that S has a limit point $\bar{x}$. From our previous result there exists a sequence of real numbers $(y_n)_{n\in\mathbb{N}}$ with $y_n \in S$ and $y_n \neq \bar{x}$ such that

$$\lim_{n\to\infty} y_n = \bar{x}.$$

Although each y_n occurs as a term in the sequence $(x_n)_{n\in\mathbb{N}}$, namely for $j \in \mathbb{N}$ $\exists\, n_j \in \mathbb{N}$ such that $y_j = x_{n_j}$, there is no reason to expect $(n_j)_{j\in\mathbb{N}}$ to be a strictly increasing sequence of real numbers. So suppose that $(n_j)_{j\in\mathbb{N}}$ has been constructed, and observe that since

$$\lim_{n\to\infty} y_n = \bar{x} \qquad (*)$$

and $y_n \neq \bar{x}$ $\forall\, n \in \mathbb{N}$, $(n_j)_{j\in\mathbb{N}}$ has an infinite number of distinct terms, for if not, then the seq $(y_n)_{n\in\mathbb{N}}$ would only have a finite number of terms and so $(*)$ can not be satisfied.

We will now show that $(n_j)_{j\in\mathbb{N}}$ has a strictly increasing seq. of real numbers $(n_{j_k})_{k\in\mathbb{N}}$.

Set $j_1 = 1$ i.e. set $n_{j_1} = n_1$. Suppose, by way of induction, that j_k, and hence n_{j_k} has been chosen.

Choose $j_{k+1} \in \mathbb{N}$ such that

$$j_{k+1} > j_k \quad \text{and} \quad n_{j_{k+1}} > n_{j_k}.$$

Note such a j_k and $n_{j_{k+1}}$ exists, since otherwise $(n_j)_{j\in\mathbb{N}}$ would only have a finite number of terms.

This completes the proof of our claim.

Thus we now have the subsequence $(x_{n_{j_k}})_{k\in\mathbb{N}}$ with

$$\lim_{k\to\infty} x_{n_{j_k}} = \lim_{k\to\infty} y_{j_k} = \lim_{j\to} y_j = \bar{x}$$

$\downarrow$

Since $(y_{j_k})_{k\in\mathbb{N}}$ is a subseq of $(y_j)_{j\in\mathbb{N}}$.

■